

Steven Lungu

+260974780588

Stevenlungu097@gmail.com

Qualifications: M. Sc Data Science | B. Sc Actuarial Science| Google Advance Data Analyst Certificate| Et. Al.

Exploratory Data Analysis Report Using the PACE Method

Brazilian E-Commerce Public Dataset by Olist

1. Project Overview

This project conducts Exploratory Data Analysis on the Brazilian E-Commerce Public Dataset by Olist. The analysis uses multiple related datasets, including orders, customers, order items, payments, reviews, products, sellers, geolocation, and product category translation tables. These datasets were loaded and merged into a consolidated analytical dataset for further exploration.

The final merged dataset contains **118,310 rows and 40 columns**, making it suitable for analysing customer behaviour, product performance, payment patterns, delivery performance, and revenue trends.

PACE Framework

P — Plan

The objective of this EDA project is to understand the structure, quality, and business meaning of the Olist e-commerce data before moving into deeper analytics or dashboard development.

Key Business Questions

The analysis is guided by the following questions:

1. What is the overall structure of the merged e-commerce dataset?
2. Which fields contain missing values, and how significant are they?
3. Are there duplicate records in the dataset?
4. What are the main product categories by order volume and revenue?
5. Which customer states contribute the highest order activity?
6. What are the most common payment methods?
7. How do orders and revenue change over time?
8. What is the typical delivery time for customer orders?

Steven Lungu

+260974780588

Stevenlungu097@gmail.com

Qualifications: M. Sc Data Science | B. Sc Actuarial Science| Google Advance Data Analyst Certificate| Et. Al.

Data Sources Used

The notebook loads the following datasets: `olist_geolocation_dataset.csv`, `olist_orders_dataset.csv`, `olist_order_reviews_dataset.csv`, `olist_order_items_dataset.csv`, `olist_customers_dataset.csv`, `olist_order_payments_dataset.csv`, `olist_products_dataset.csv`, `product_category_name_translation.csv`, and `olist_sellers_dataset.csv`.

Data Integration Approach

The datasets were merged using common keys. The `orders` table connects to `customers` through `customer_id`, while `orders`, `order_items`, `payments`, and `reviews` are connected through `order_id`. The `order_items` table connects to `products` through `product_id` and to `sellers` through `seller_id`. The `products` table connects to the category translation table through `product_category_name`.

A — Analyse

The analysis phase focused on understanding the dataset structure, identifying data quality issues, and generating initial descriptive statistics.

Dataset Structure

After merging, the dataset contains 118,310 observations and 40 variables. The dataframe includes order identifiers, product identifiers, seller identifiers, customer details, product attributes, payment information, review information, and order delivery dates.

The merged dataset initially contained 24 object columns, 13 float columns, and 3 integer columns before date conversion. After converting `order_purchase_timestamp` and `order_delivered_customer_date`, the dataset contained 2 datetime columns, 13 float columns, 3 integer columns, and 22 object columns.

Steven Lungu

+260974780588

Stevenlungu097@gmail.com

Qualifications: M. Sc Data Science | B. Sc Actuarial Science| Google Advance Data Analyst Certificate| Et. Al.

Missing Values Analysis

The missing values analysis identified several areas requiring attention. The most significant issue is in customer review text fields. `review_comment_title` has 104,418 missing values, representing 88.26% of the dataset. `review_comment_message` has 68,628 missing values, representing 58.01% of the dataset.

Delivery-related fields also contain missing values. `order_delivered_customer_date` has 2,588 missing values, representing 2.19%, while `order_delivered_carrier_date` has 1,254 missing values, representing 1.06%.

Product classification fields also have missing values. `product_category_name` has 1,709 missing values, while `product_category_name_english` has 1,734 missing values. These represent approximately 1.44% and 1.47% of the dataset respectively.

Payment-related missing values are minimal. `payment_type`, `payment_value`, `payment_installments`, and `payment_sequential` each have 3 missing values, representing 0.003% of the dataset.

Duplicate Records

The duplicate check returned **0 duplicate rows**, indicating that the merged dataset does not contain fully duplicated records.

Descriptive Statistics

The average item price is 120.65, while the median item price is 74.90. The minimum price is 0.85, and the maximum price is 6,735.00, indicating a right-skewed price distribution with a small number of high-value transactions.

The average freight value is 20.03, with a median of 16.28. Freight values range from 0.00 to 409.68.

The order purchase timestamps range from 4 September 2016 to 3 September 2018, while delivered customer dates range from 11 October 2016 to 17 October 2018.

C — Construct

Steven Lungu

+260974780588

Stevenlungu097@gmail.com

Qualifications: M. Sc Data Science | B. Sc Actuarial Science| Google Advance Data Analyst Certificate| Et. Al.

The construction phase focused on preparing analytical features and creating useful aggregations for business interpretation.

Date Conversion

The `order_purchase_timestamp` and `order_delivered_customer_date` columns were converted from object format into datetime format. This enabled time-series analysis and delivery time calculations.

Time-Based Feature Creation

A `year_month` variable was created from `order_purchase_timestamp` to support monthly order and revenue trend analysis.

Delivery Time Feature

A new delivery time variable was created by subtracting `order_purchase_timestamp` from `order_delivered_customer_date`. The resulting delivery time was measured in days.

The delivery time analysis shows 115,722 valid delivery records, with an average delivery time of 12.02 days, a median of 10 days, and a standard deviation of 9.45 days. The minimum delivery time is 0 days.

Category-Level Aggregation

Product category analysis was performed using `product_category_name_english`. The top product categories by order count include `bed_bath_table`, `health_beauty`, `sports_leisure`, `furniture_decor`, and `computers_accessories`.

The top category by order count is `bed_bath_table` with 11,988 records, followed by `health_beauty` with 10,032 records, `sports_leisure` with 9,004 records, `furniture_decor` with 8,832 records, and `computers_accessories` with 8,150 records.

Steven Lungu
+260974780588

Stevenlungu097@gmail.com

Qualifications: M. Sc Data Science | B. Sc Actuarial Science| Google Advance Data Analyst Certificate| Et. Al.

Revenue Aggregation

Revenue by category was calculated using the `price` field. The top revenue-generating product category is `health_beauty` with 1,301,947.97, followed by `watches_gifts` with 1,254,322.95, `bed_bath_table` with 1,107,249.09, `sports_leisure` with 1,029,603.88, and `computers_accessories` with 950,053.69.

E — Execute

The execution phase focused on interpreting the analytical outputs and presenting business insights.

Key Finding 1: Dataset Is Suitable for Transactional E-Commerce Analysis

The merged dataset provides a broad view of e-commerce operations, covering order activity, customers, products, sellers, payments, reviews, and delivery information. With 118,310 rows and 40 columns, the dataset provides sufficient depth for further analysis and dashboard development.

Key Finding 2: Review Comments Have High Missingness

The review comment fields contain substantial missing values. `review_comment_title` is missing in 88.26% of records, while `review_comment_message` is missing in 58.01% of records. This suggests that many customers provide numeric review scores without leaving detailed written feedback.

Key Finding 3: No Fully Duplicate Rows Were Found

The duplicate check returned **0 duplicated rows**, indicating that the final merged dataset does not contain exact full-row duplication.

Key Finding 4: Product Prices Are Right-Skewed

The average price is 120.65, while the median is 74.90. Since the mean is higher than the median and the maximum price is 6,735.00, the price distribution appears to be influenced by high-value transactions.

Key Finding 5: Health and Beauty Generates the Highest Revenue

Steven Lungu

+260974780588

Stevenlungu097@gmail.com

Qualifications: M. Sc Data Science | B. Sc Actuarial Science| Google Advance Data Analyst Certificate| Et. Al.

Although `bed_bath_table` has the highest order count, `health_beauty` generates the highest revenue at 1,301,947.97. This indicates that the highest-volume category is not necessarily the highest-revenue category.

Key Finding 6: São Paulo Dominates Customer Activity

The customer state with the highest number of records is `SP`, with 49,865 records. Other high-volume states include `RJ` with 15,425 records and `MG` with 13,718 records.

Key Finding 7: Credit Card Is the Most Common Payment Method

Credit card is the dominant payment method, with 87,258 records. This is followed by boleto with 23,018 records, voucher with 6,332 records, and debit card with 1,699 records.

Key Finding 8: Average Delivery Time Is Approximately 12 Days

The average delivery time is 12.02 days, with a median of 10 days. This suggests that most orders are delivered within approximately one to two weeks, based on the available delivery records.

Recommendations

1. Treat Review Text Fields Separately

Because `review_comment_title` and `review_comment_message` have high missingness, these fields should not be used as mandatory variables in modelling or dashboard KPIs. They can be analysed separately for text analytics where comments are available.

2. Use `price` Carefully as the Revenue Measure

The current revenue analysis uses `price`. If a broader revenue measure is required, a new variable combining `price` and `freight_value` may be created depending on whether freight should be treated as part of customer-paid revenue or logistics cost.

3. Address Missing Product Categories

Steven Lungu

+260974780588

Stevenlungu097@gmail.com

Qualifications: M. Sc Data Science | B. Sc Actuarial Science| Google Advance Data Analyst Certificate| Et. Al.

Missing values in `product_category_name` and `product_category_name_english` should be cleaned or classified as `Unknown` to avoid losing observations during category-level analysis.

4. Use Unique Order Counts for Order-Level Metrics

Since the merged dataset is built from order items and includes payments and reviews, care should be taken when counting orders. For order-level KPIs, `order_id.nunique()` should be used instead of simple row counts.

5. Expand the Delivery Analysis

Delivery time analysis should be extended by comparing actual delivery dates against estimated delivery dates. This would support an on-time delivery performance KPI and provide stronger operational insight.

Conclusion

The EDA shows that the Olist e-commerce dataset is rich, well-structured, and suitable for analysing sales, customer geography, products, payments, reviews, and logistics performance. The dataset contains 118,310 rows and 40 columns, with no fully duplicated records. Key data quality concerns include high missingness in review comment fields, some missing delivery dates, and missing product category labels. From the initial analysis, `health_beauty` is the leading revenue category, `bed_bath_table` is the most frequent product category, `SP` is the dominant customer state, and credit card is the most common payment method.

Overall, this project provides a strong foundation for further analysis, including customer segmentation, sales trend analysis, delivery performance monitoring, and Power BI dashboard development.